Shower Shape Comparison: Nominal vs Double-Interaction Mixed MC

0 mrad (22.4% DI) and 1.5 mrad (7.9% DI) Crossing Angles

PPG12 Analysis Team (automated report)

April 15, 2026 (regenerated with iterative truth-vertex reweight)

1 Introduction

In *pp* collisions at RHIC, two independent collisions can occur in the same bunch crossing (double interaction / pileup), distorting reconstructed cluster kinematics and shower shapes. To quantify this effect, full GEANT4 double-interaction MC samples (`photon10_double`, `jet12_double`) are blended with nominal single-interaction MC at the physics-dictated fractions:

- **0 mrad** crossing angle (runs 47289–51274, $\mathcal{L} = 32.7$ pb^{-1}): 22.4% double interaction (cluster-weighted)
- **1.5 mrad** crossing angle (runs 51274–54000, $\mathcal{L} = 16.9$ pb^{-1}): 7.9% double interaction (cluster-weighted)

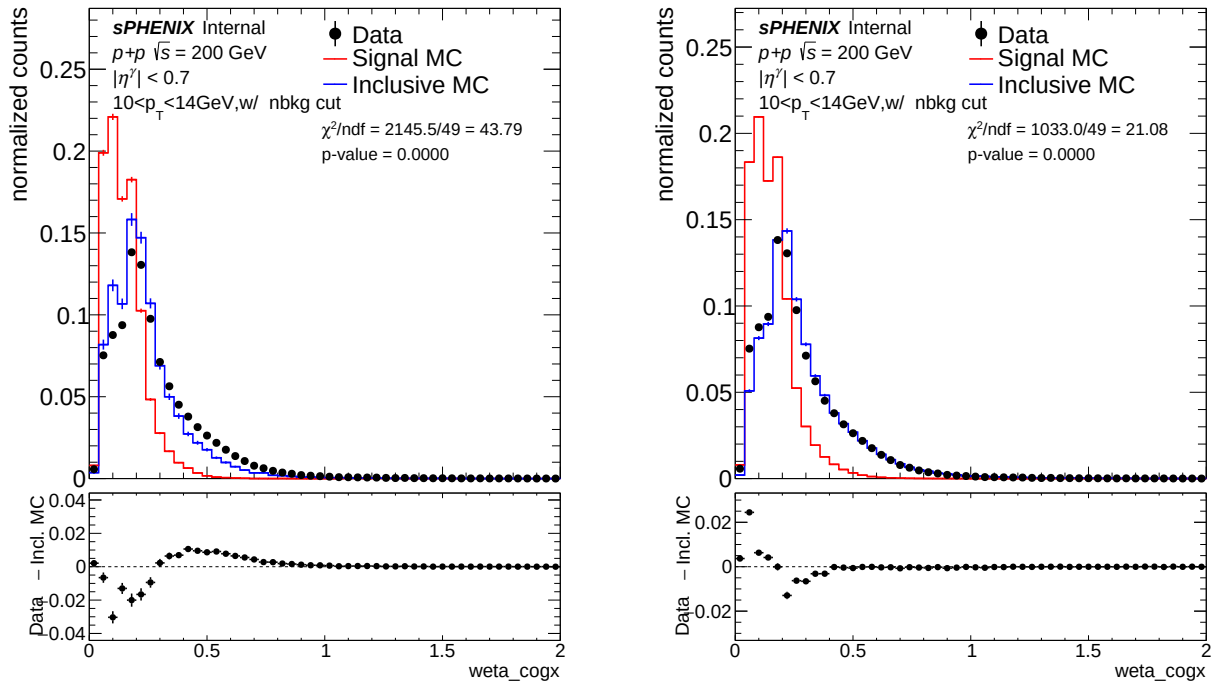
The blending uses a two-pass pipeline (`run_showershape_double.sh`): Pass 1 builds a combined vertex distribution (weighted `hadd` of single and double vtxscan files); Pass 2 reruns the full shower shape analysis with `mix_weight` applied to all histogram fills. The final "mixed MC" is the `hadd` of single-weighted and double-weighted output files. MC vertex reweighting uses the *iterative truth-vertex* method (applied per truth vertex, $w(z_h)$ for single MC, $w(z_h)\,w(z_{\mathrm{mb}})$ for double MC; see `truth_vertex_reco_check` report) rather than the earlier reco-level $f(z_r)$ ratio.

All comparisons use the **cut1** selection level (with NPB preselection). Each plot shows data (black points), signal MC (red), and inclusive MC (blue), area-normalized, with a χ^2/ndf and p-value annotation and a Data − Inclusive MC difference panel. Left panels show nominal (single-only) MC; right panels show mixed (double-blended) MC.

2 0 mrad Crossing Angle (22.4% Double Interaction)

2.1 w_η^{CoG} (Eta Cluster Width)

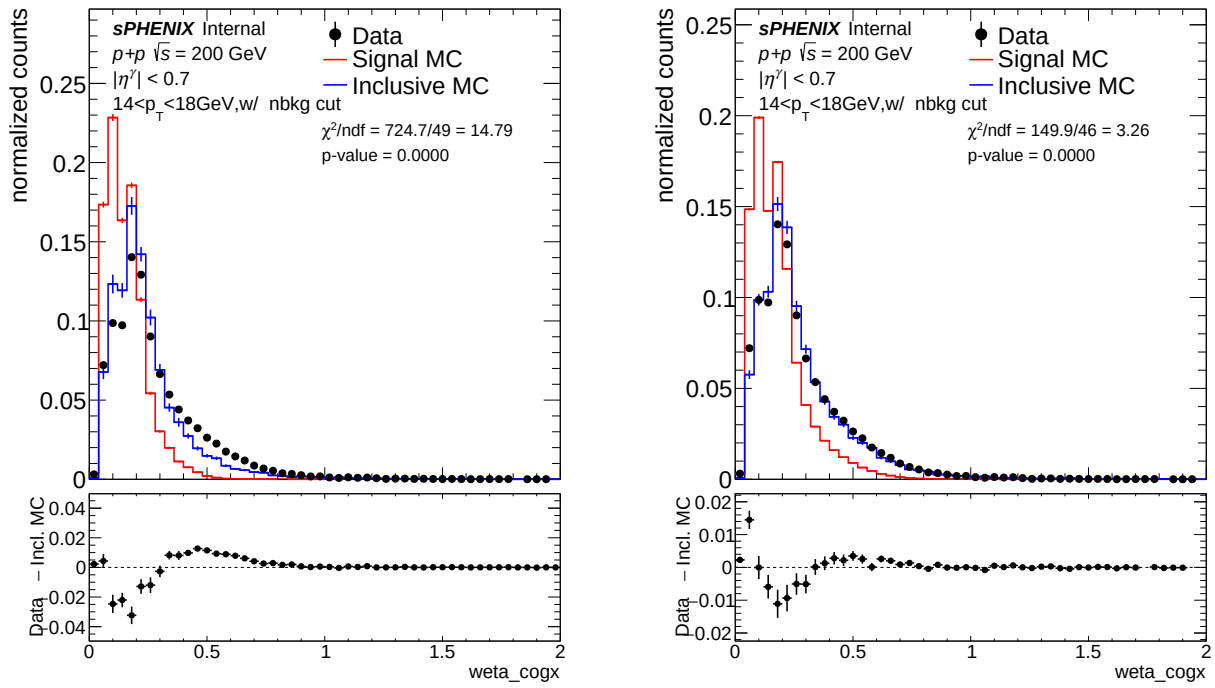
w_η^{CoG} is strongly affected by double interactions: vertex shifts displace clusters along the beam axis, broadening the η profile.



(a) Nominal MC

(b) Mixed MC (22.4% DI)

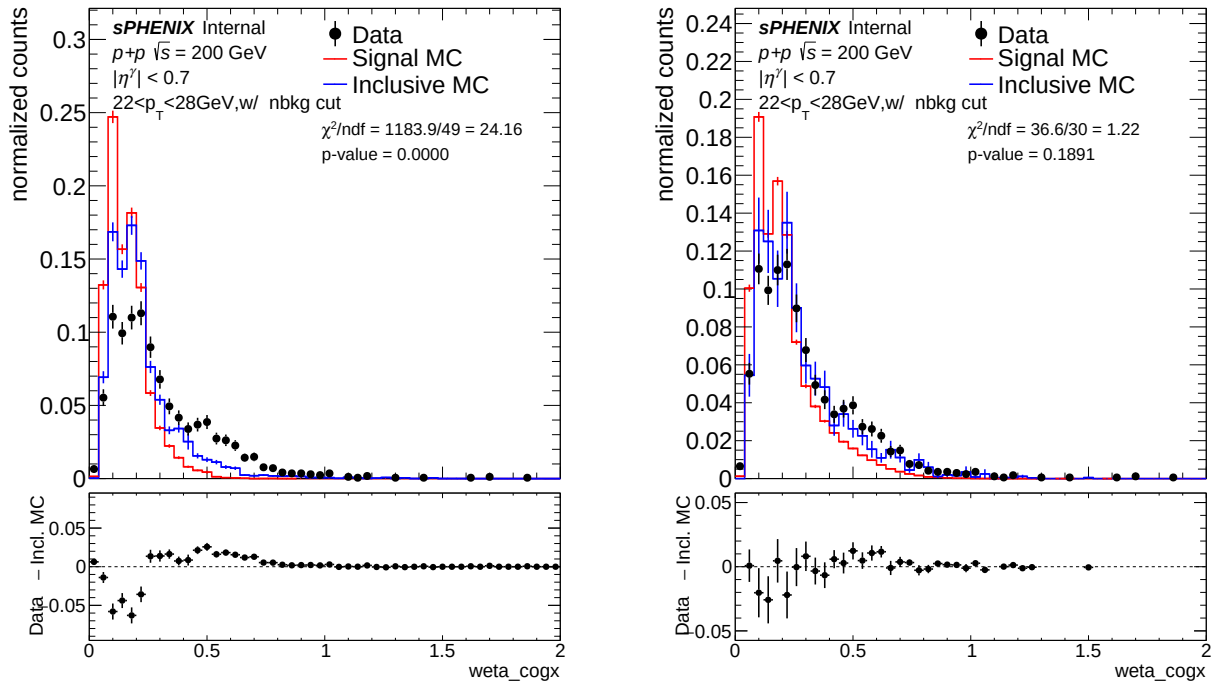
Figure 1: w_η^{CoG}, $10 < E_T < 14\,\mathrm{GeV}$ (pt0), $0\,\mathrm{mrad}$. Nominal $\chi^2/\mathrm{ndf} = 47.40$; mixed $\chi^2/\mathrm{ndf} = 28.96$ ($1.6\times$ improvement).



(a) Nominal MC

(b) Mixed MC (22.4% DI)

Figure 2: w_η^{CoG}, $14 < E_T < 18\,\mathrm{GeV}$ (pt1), $0\,\mathrm{mrad}$. Nominal $\chi^2/\mathrm{ndf} = 17.64$; mixed $\chi^2/\mathrm{ndf} = 2.99$ ($5.9\times$ improvement).



(a) Nominal MC

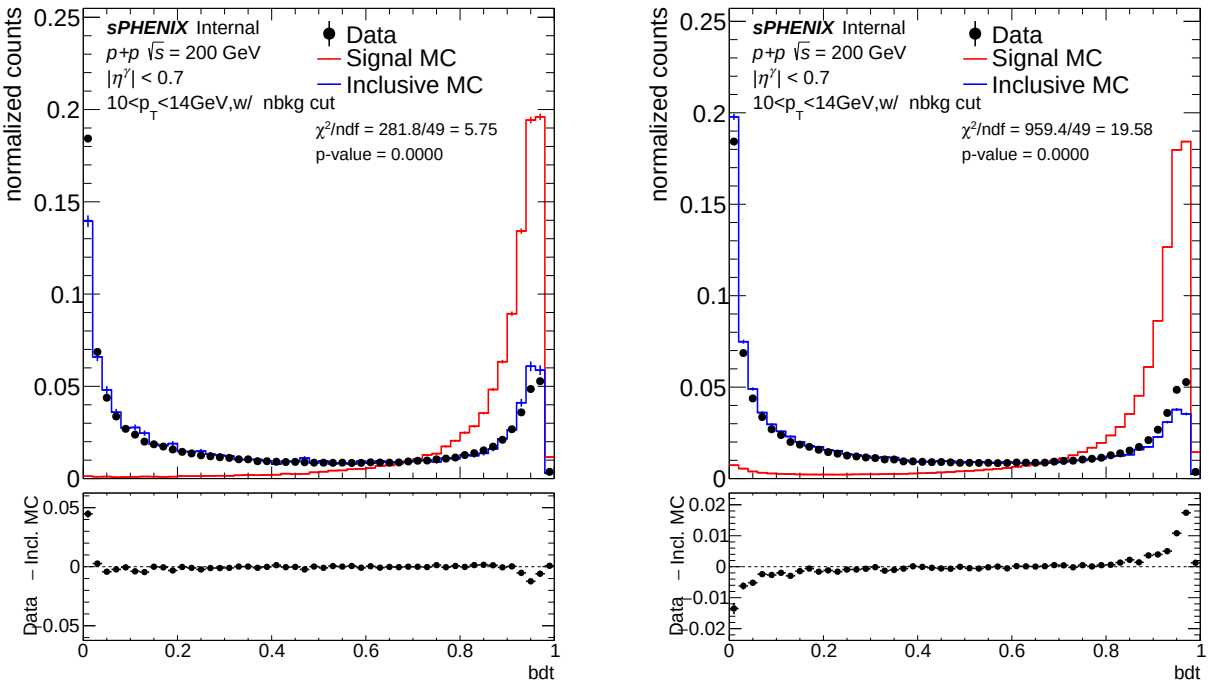
(b) Mixed MC (22.4% DI)

Figure 3: w_η^{CoG}, $22 < E_T < 28\,\mathrm{GeV}$ (pt3), 0 mrad. Nominal $\chi^2/\mathrm{ndf} = 15.58$; mixed $\chi^2/\mathrm{ndf} = 0.75$ ($p = 0.84$; 20.8× improvement).

The mixed MC improves data–MC agreement for w_η^{CoG} across all E_T bins at 0 mrad, with χ^2/ndf dropping by factors of 1.6 (pt0) to 20.8 (pt3). At pt3, the mixed MC achieves $p = 0.84$.

2.2 BDT Score

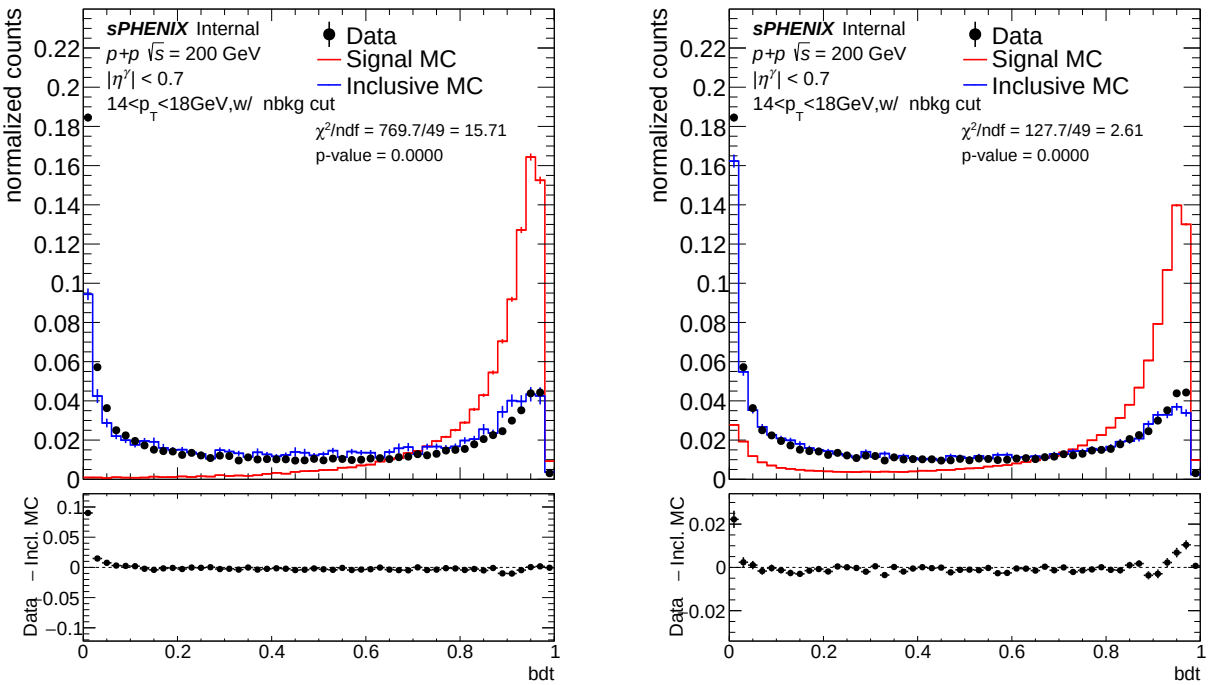
The photon/jet BDT discriminator amplifies underlying shower shape distortions through the XGBoost decision function.



(a) Nominal MC

(b) Mixed MC (22.4% DI)

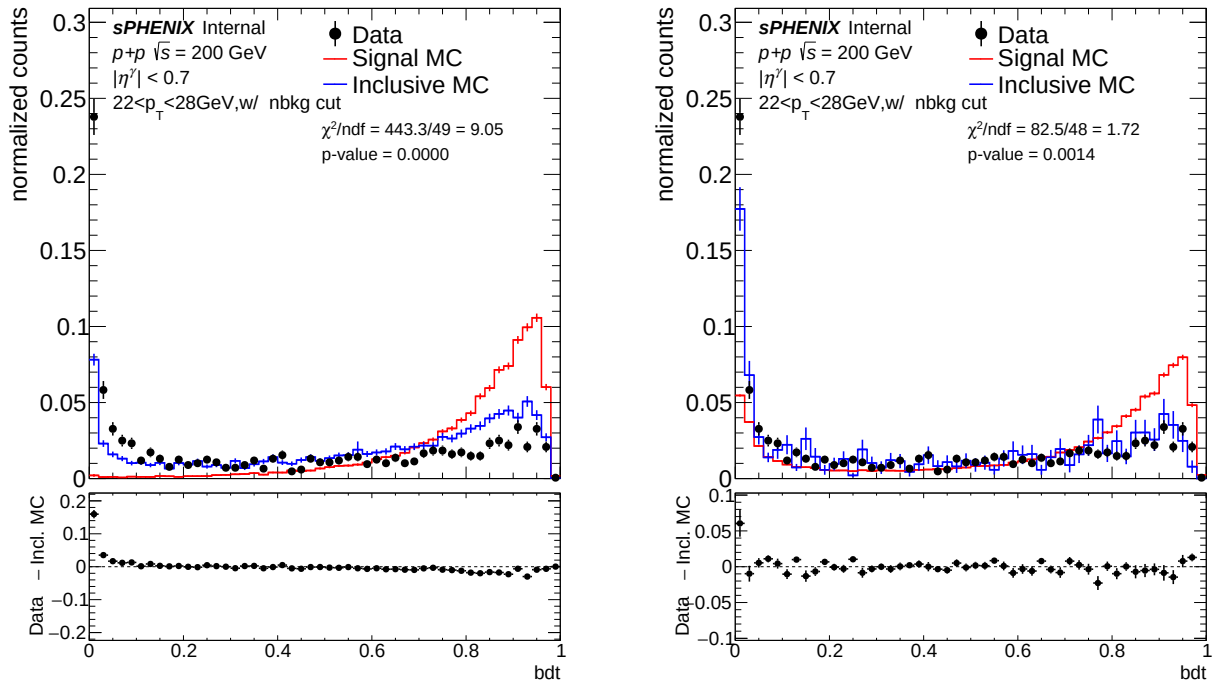
Figure 4: BDT score, $10 < E_T < 14\,\text{GeV}$ (pt0), 0 mrad. Nominal $\chi^2/\text{ndf} = 6.85$; mixed $\chi^2/\text{ndf} =$ 20.30 (3.0× **degradation**).



(a) Nominal MC

(b) Mixed MC (22.4% DI)

Figure 5: BDT score, $14 < E_T < 18\,\text{GeV}$ (pt1), 0 mrad. Nominal $\chi^2/\text{ndf} = 12.20$; mixed $\chi^2/\text{ndf} =$ 1.63 ($p = 0.004$; 7.5× improvement).



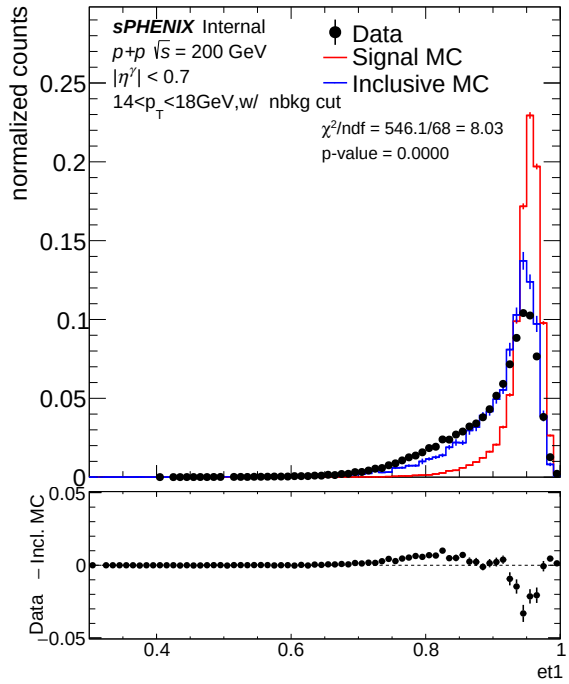
(a) Nominal MC

(b) Mixed MC (22.4% DI)

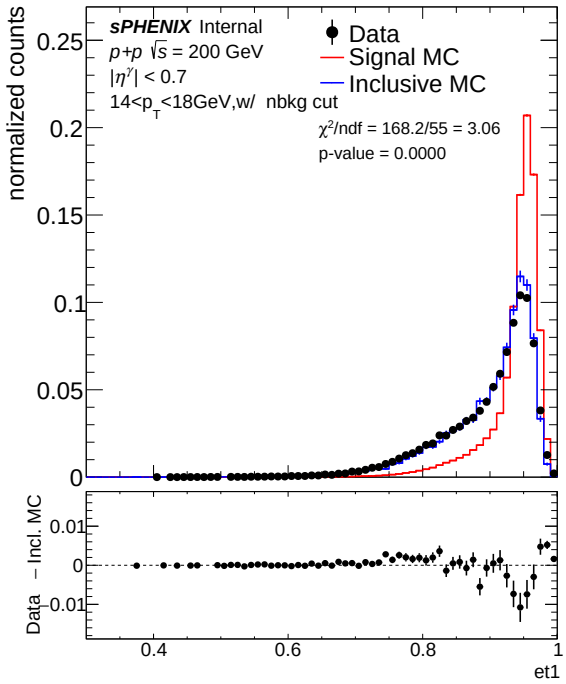
Figure 6: BDT score, $22 < E_T < 28\,\text{GeV}$ (pt3), 0 mrad. Nominal $\chi^2/\text{ndf} = 9.62$; mixed $\chi^2/\text{ndf} = 1.75$ ($p = 0.001$; 5.5× improvement).

At 0 mrad, the BDT χ^2/ndf improves substantially at pt1 (7.5×) and pt3 (5.5×). However, at pt0, the mixed MC *degrades* agreement by 3.0× (6.85 → 20.30). This low-E_T degradation was not reported in the original 0 mrad analysis and is now understood to be a common feature of both crossing angles (see Section 6).

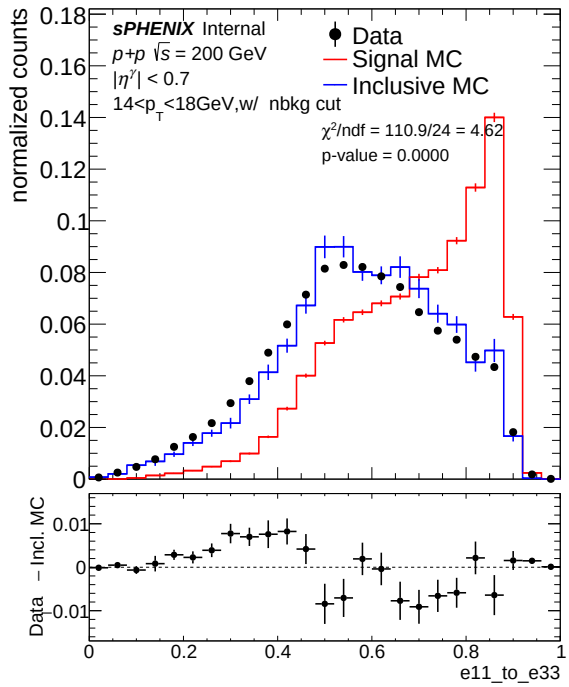
2.3 f_1 and $E_{1\times1}/E_{3\times3}$



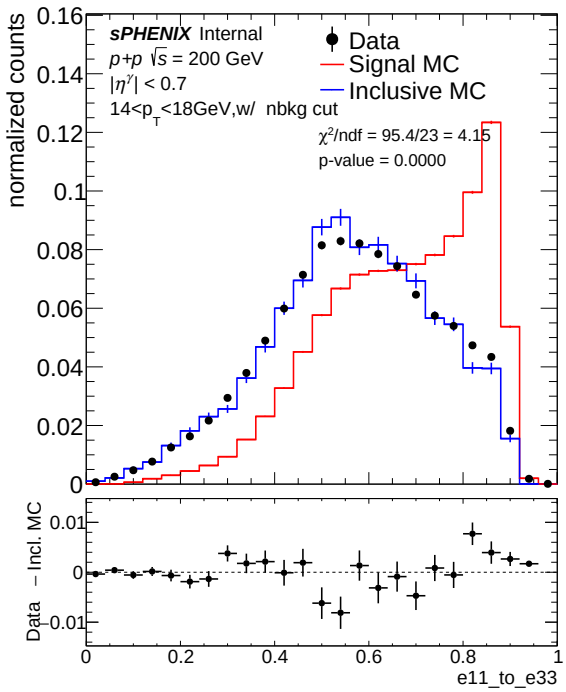
(a) f_1, Nominal MC



(b) f_1, Mixed MC (22.4% DI)



(c) $E_{1\times1}/E_{3\times3}$, Nominal MC

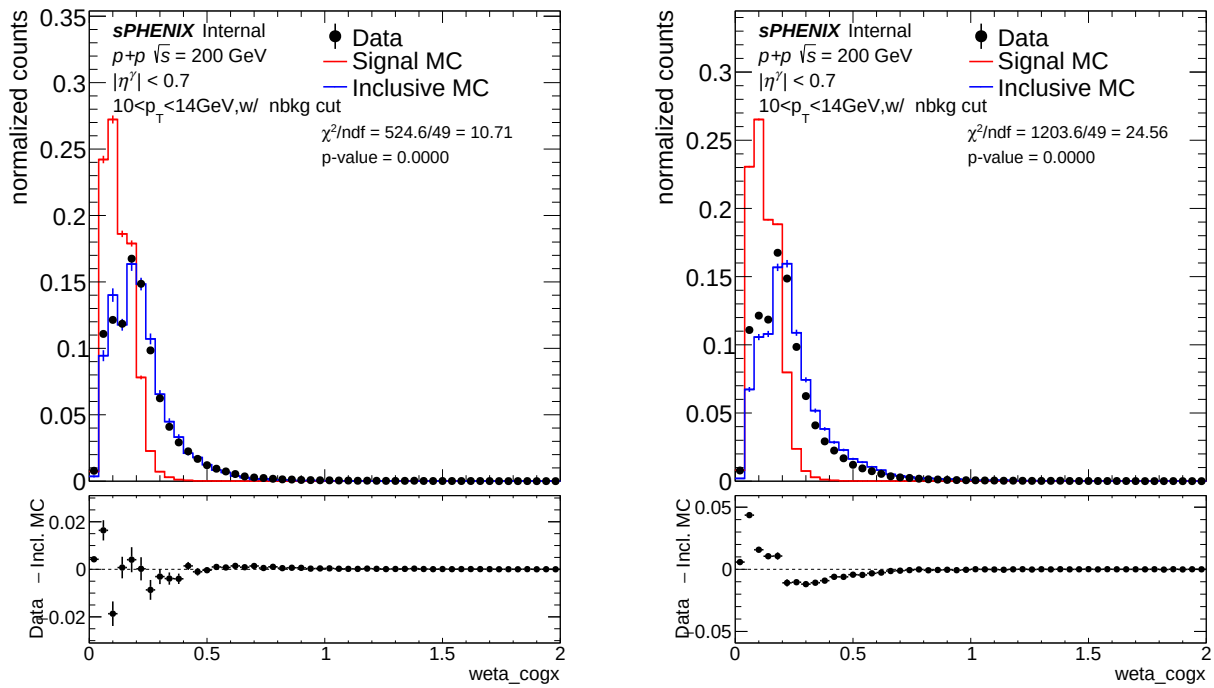


(d) $E_{1\times1}/E_{3\times3}$, Mixed MC (22.4% DI)

Figure 7: f_1 (top) and $E_{1\times1}/E_{3\times3}$ (bottom) at $14 < E_T < 18\,\text{GeV}$ (pt1), 0 mrad. f_1: $\chi^2/\text{ndf} = 9.33 \rightarrow 1.92$ (4.9×). $E_{1\times1}/E_{3\times3}$: $\chi^2/\text{ndf} = 4.50 \rightarrow 4.14$ (1.1×).

3 1.5 mrad Crossing Angle (7.9% Double Interaction)

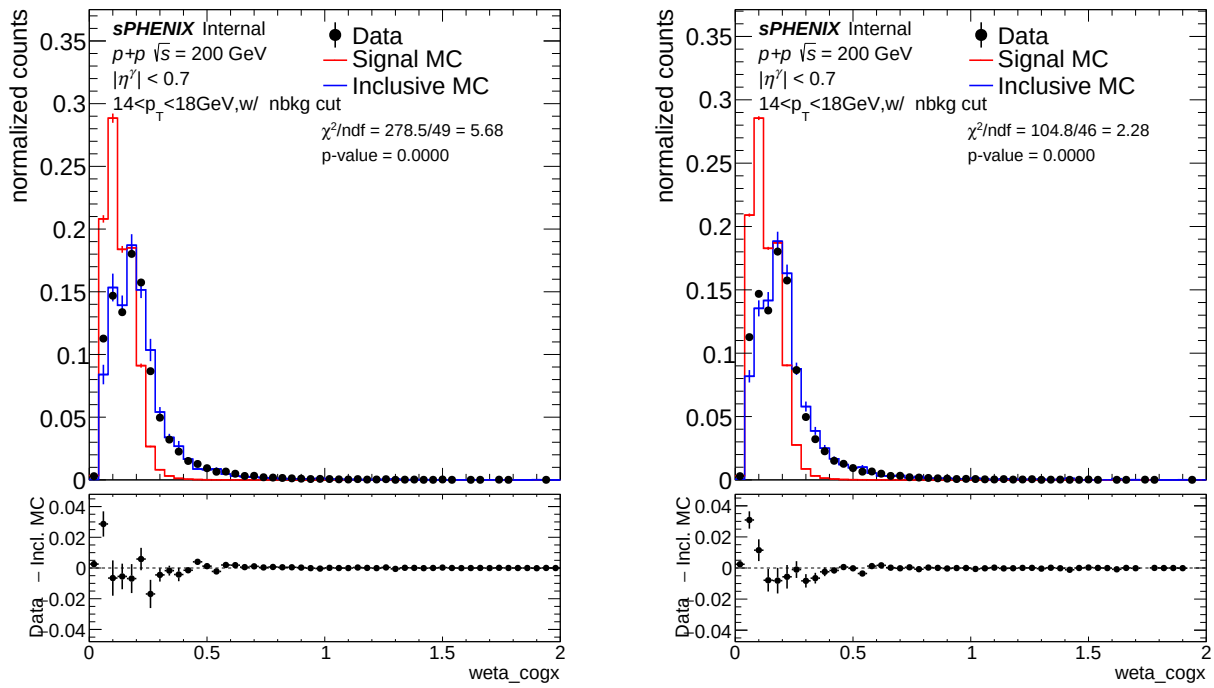
3.1 w_η^{CoG} (Eta Cluster Width)



(a) Nominal MC

(b) Mixed MC (7.9% DI)

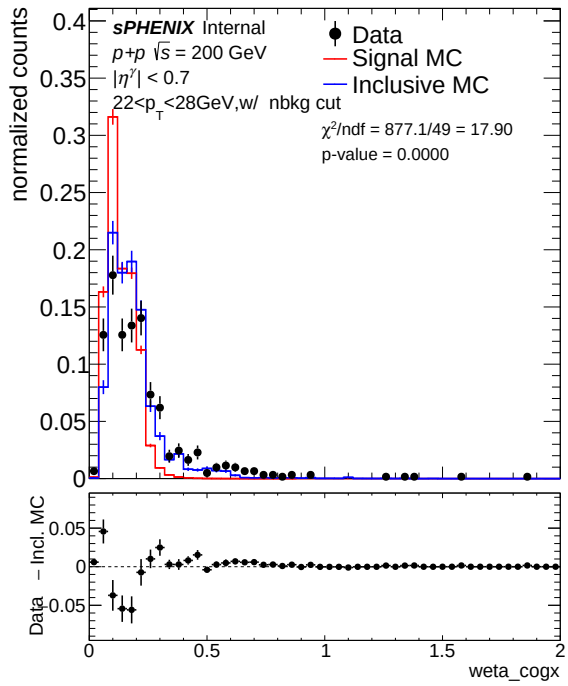
Figure 8: w_η^{CoG}, $10 < E_T < 14\,\mathrm{GeV}$ (pt0), 1.5 mrad. Nominal $\chi^2/\mathrm{ndf} = 9.72$; mixed $\chi^2/\mathrm{ndf} = 42.31$ (4.4× **degradation**).



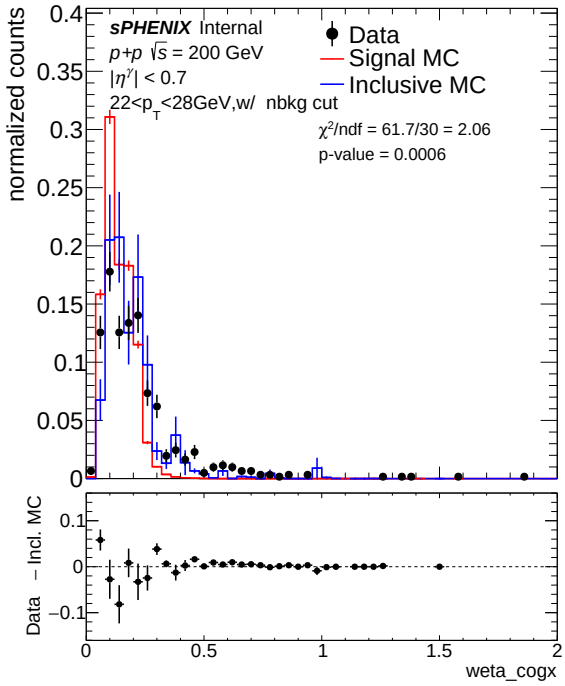
(a) Nominal MC

(b) Mixed MC (7.9% DI)

Figure 9: w_η^{CoG}, $14 < E_T < 18\,\mathrm{GeV}$ (pt1), 1.5 mrad. Nominal $\chi^2/\mathrm{ndf} = 6.73$; mixed $\chi^2/\mathrm{ndf} = 4.86$ (1.4× improvement).



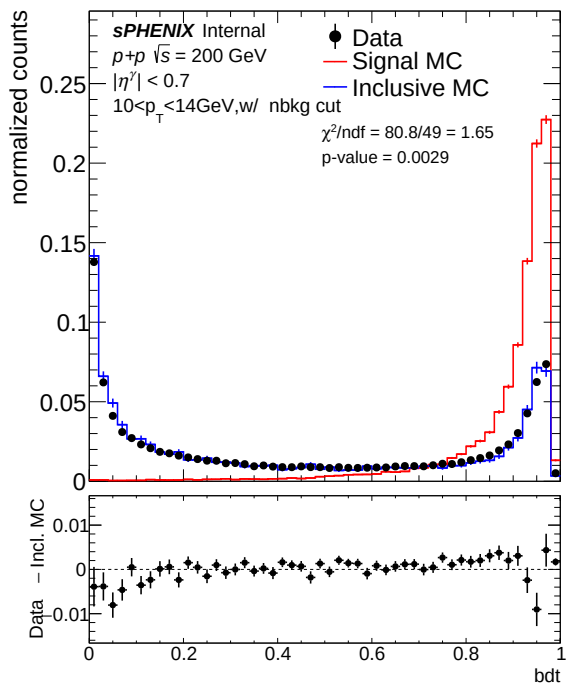
(a) Nominal MC



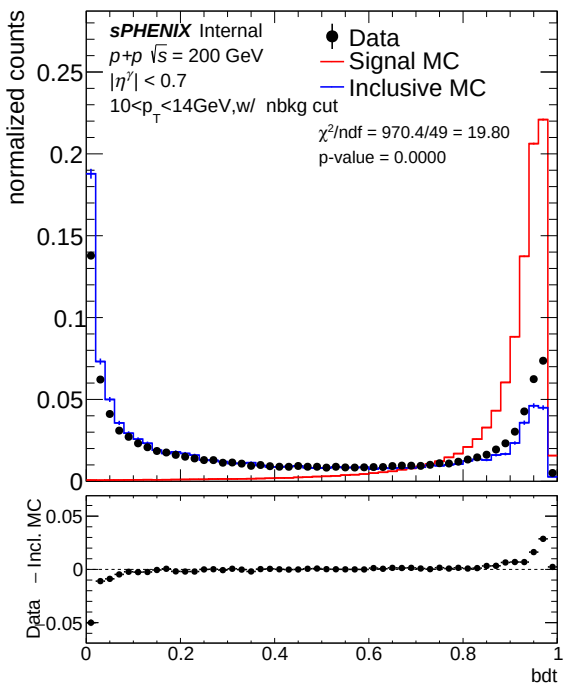
(b) Mixed MC (7.9% DI)

Figure 10: w_η^{CoG}, $22 < E_T < 28\,\mathrm{GeV}$ (pt3), 1.5 mrad. Nominal $\chi^2/\mathrm{ndf} = 21.10$; mixed $\chi^2/\mathrm{ndf} = 1.31$ ($p = 0.118$; 16.1× improvement).

3.2 BDT Score

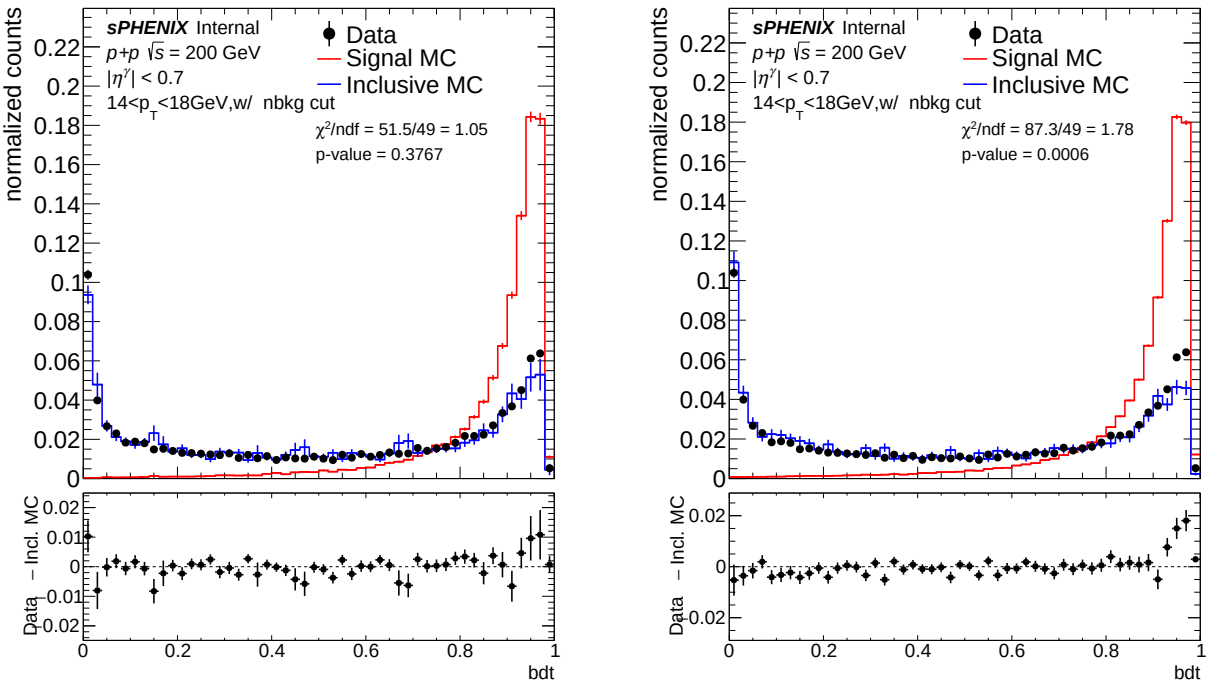


(a) Nominal MC



(b) Mixed MC (7.9% DI)

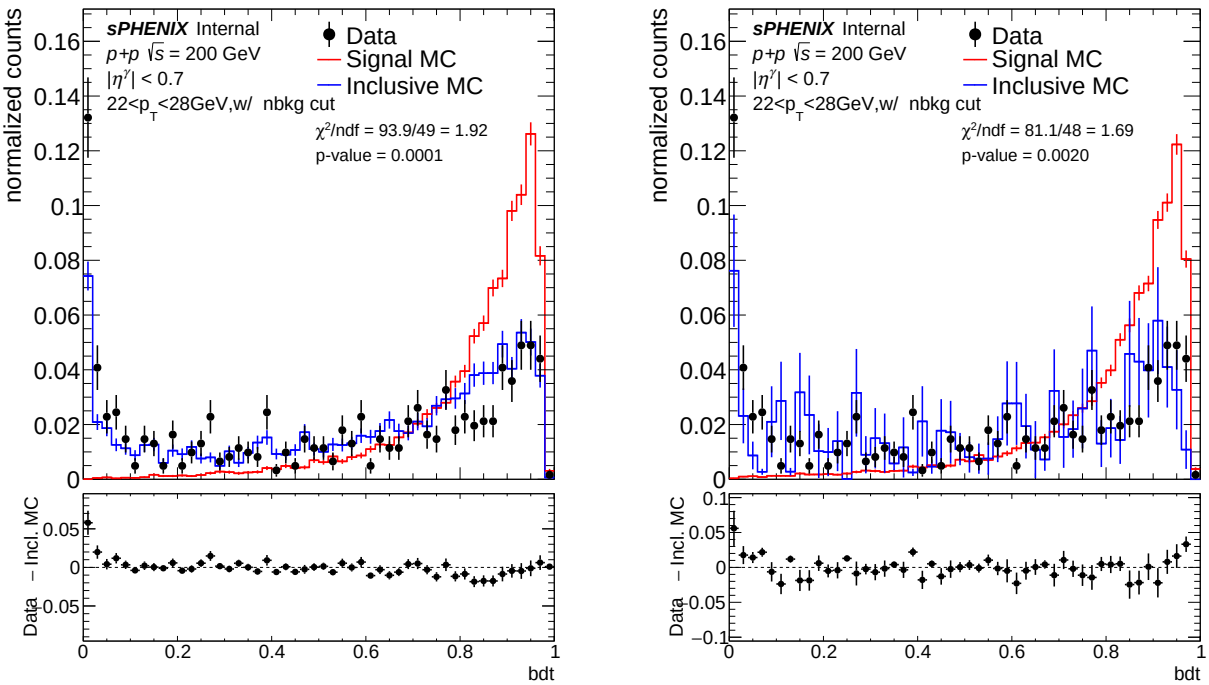
Figure 11: BDT score, $10 < E_T < 14\,\mathrm{GeV}$ (pt0), 1.5 mrad. Nominal $\chi^2/\mathrm{ndf} = 5.98$; mixed $\chi^2/\mathrm{ndf} = 33.08$ (5.5× **degradation**).



(a) Nominal MC

(b) Mixed MC (7.9% DI)

Figure 12: BDT score, $14 < E_T < 18\,\text{GeV}$ (pt1), 1.5 mrad. Nominal $\chi^2/\text{ndf} = 1.16$ ($p = 0.207$); mixed $\chi^2/\text{ndf} = 2.31$ (2.0× **degradation**; nominal already excellent).

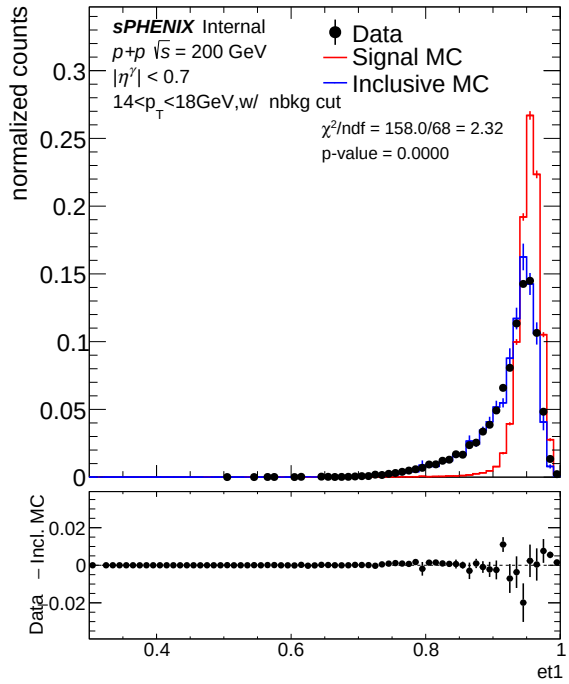


(a) Nominal MC

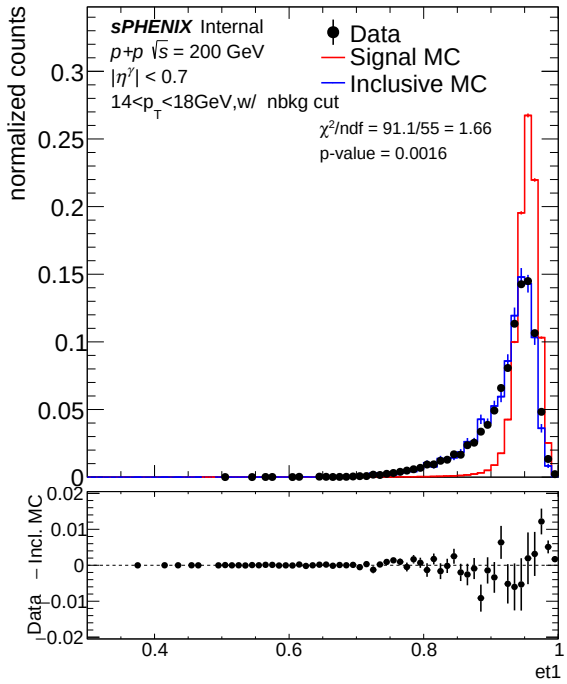
(b) Mixed MC (7.9% DI)

Figure 13: BDT score, $22 < E_T < 28\,\text{GeV}$ (pt3), 1.5 mrad. Nominal $\chi^2/\text{ndf} = 1.89$; mixed $\chi^2/\text{ndf} = 1.13$ ($p = 0.249$; 1.7× improvement).

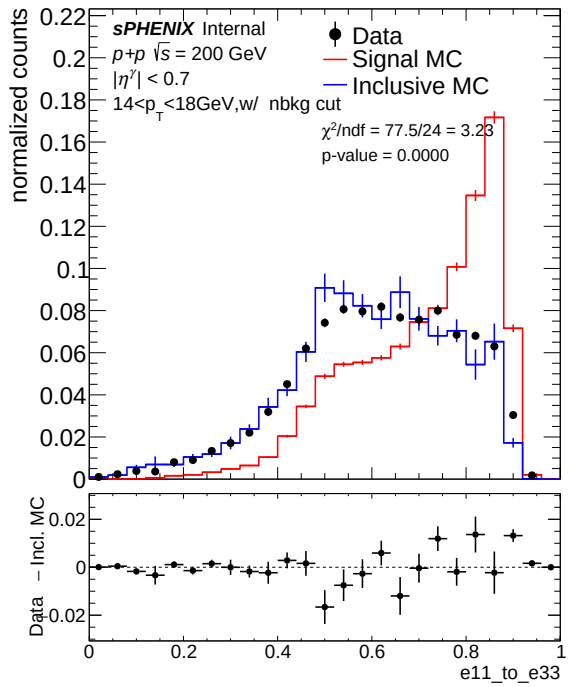
3.3 f_1 and $E_{1\times1}/E_{3\times3}$



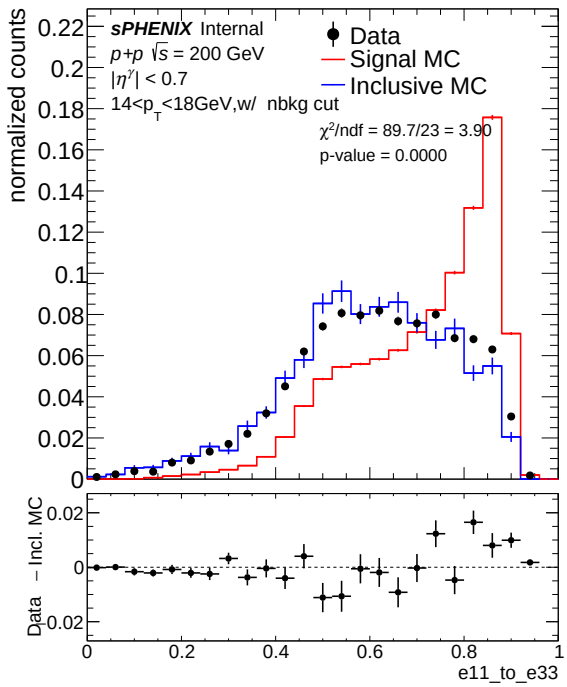
(a) f_1, Nominal MC



(b) f_1, Mixed MC (7.9% DI)



(c) $E_{1\times1}/E_{3\times3}$, Nominal MC

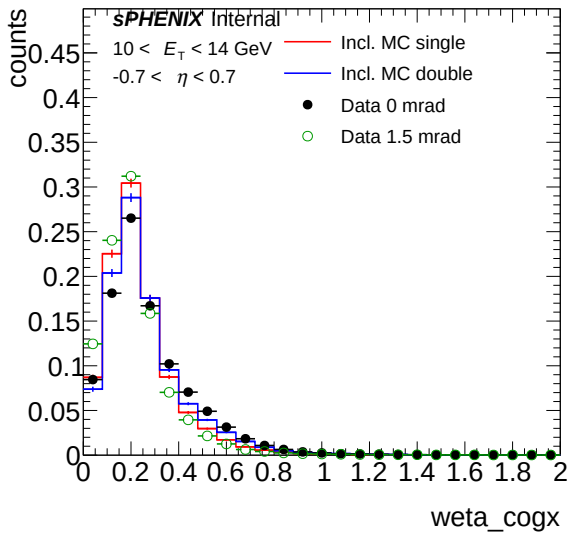


(d) $E_{1\times1}/E_{3\times3}$, Mixed MC (7.9% DI)

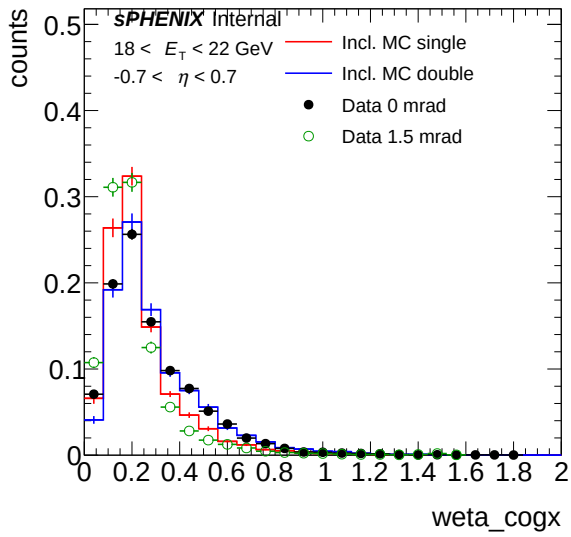
Figure 14: f_1 (top) and $E_{1\times1}/E_{3\times3}$ (bottom) at $14 < E_T < 18\,\text{GeV}$ (pt1), 1.5 mrad. f_1: $\chi^2/\text{ndf} = 4.17 \rightarrow 2.55$ (1.6×). $E_{1\times1}/E_{3\times3}$: $\chi^2/\text{ndf} = 6.88 \rightarrow 6.04$ (1.1×).

4 Single vs Double Interaction Overlay

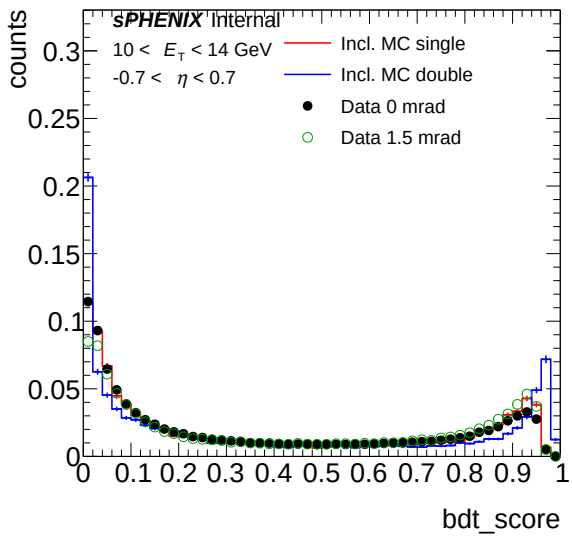
Figure 15 overlays single-interaction MC (red) and double-interaction MC (blue) with data from both crossing-angle periods: 0 mrad (filled black circles) and 1.5 mrad (open green circles), all area-normalized. Data at 0 mrad shows larger deviation from single MC, consistent with the higher pileup fraction, while data at 1.5 mrad tracks closer to single MC.



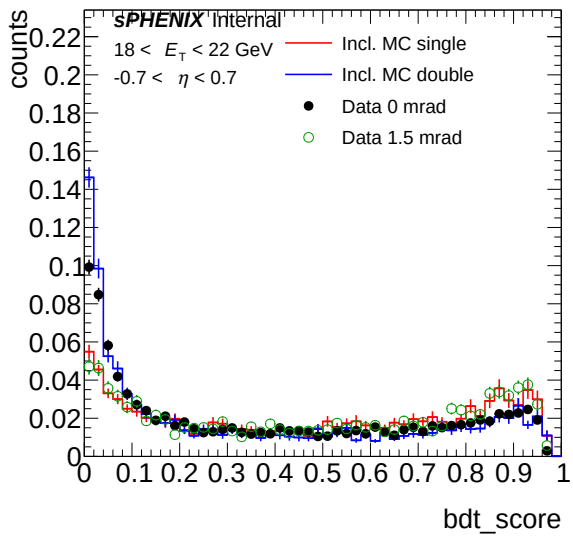
(a) w_η^{CoG}, 10–14 GeV



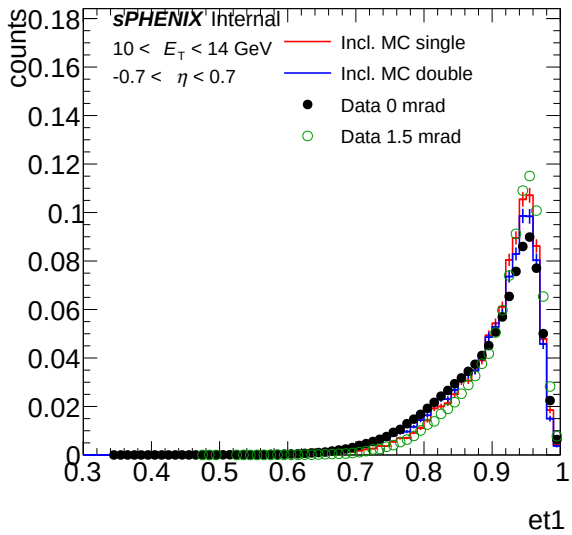
(b) w_η^{CoG}, 18–22 GeV



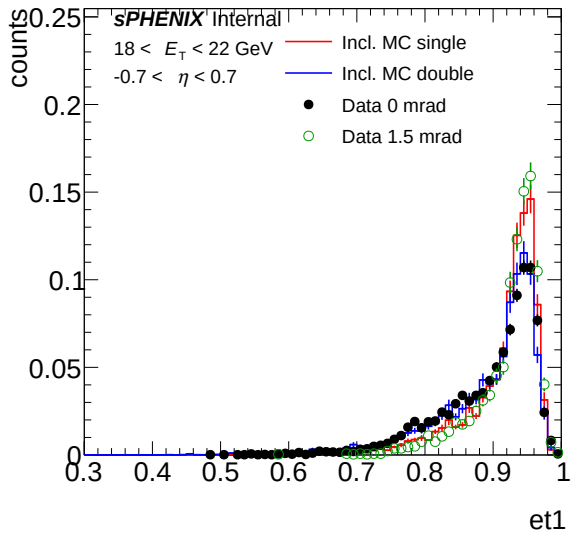
(c) BDT score, 10–14 GeV



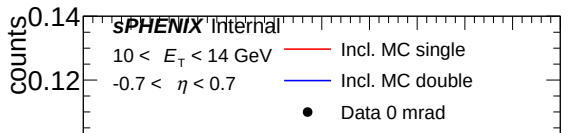
(d) BDT score, 18–22 GeV

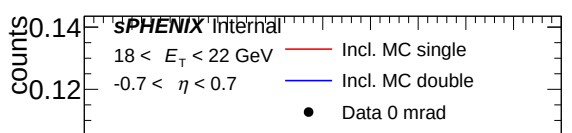


(e) f_1, 10–14 GeV



(f) f_1, 18–22 GeV





5 χ^2/ndf Summary

Table 1 compiles the Data–Inclusive MC χ^2/ndf and p-values for all variables, E_T bins, and crossing angles. Entries where the mixed MC *degrades* agreement relative to nominal are highlighted in red. Improvement factors are computed as Nominal/Mixed when the mixed MC improves, and Mixed/Nominal (prefixed with a minus sign) when it degrades.

Table 1: Data–Inclusive MC χ^2/ndf and p-values at cut1 for nominal (single-only) and mixed (double-blended) MC. Lower χ^2/ndf is better; p-values $\ll 1$ indicate significant data–MC disagreement.

		0 mrad (22.4% DI)				**1.5 mrad (7.9% DI)**			
Variable	E_T [GeV]	Nom.	p	Mixed	p	Nom.	p	Mixed	p
w_η^{CoG}	10–14 (pt0)	47.40	<0.001	28.96	<0.001	9.72	<0.001	42.31	<0.001
w_η^{CoG}	14–18 (pt1)	17.64	<0.001	2.99	<0.001	6.73	<0.001	4.86	<0.001
w_η^{CoG}	22–28 (pt3)	15.58	<0.001	0.75	0.840	21.10	<0.001	1.31	0.118
BDT score	10–14 (pt0)	6.85	<0.001	20.30	<0.001	5.98	<0.001	33.08	<0.001
BDT score	14–18 (pt1)	12.20	<0.001	1.63	0.004	1.16	0.207	2.31	<0.001
BDT score	22–28 (pt3)	9.62	<0.001	1.75	0.001	1.89	<0.001	1.13	0.249
f_1	14–18 (pt1)	9.33	<0.001	1.92	<0.001	4.17	<0.001	2.55	<0.001
$E_{1\times1}/E_{3\times3}$	14–18 (pt1)	4.50	<0.001	4.14	<0.001	6.88	<0.001	6.04	<0.001

6 Discussion

6.1 Universal pattern: E_T-dependent crossover

A consistent pattern emerges across *both* crossing angles: the double-interaction blending **degrades** data–MC agreement for the BDT score at low E_T (pt0, 10–14 GeV) while **improving** it at higher E_T. This is now confirmed for both 0 mrad (3.0× BDT degradation at pt0) and 1.5 mrad (5.5× BDT degradation at pt0). For w_η^{CoG}, the crossover occurs between pt0 and pt1: at 0 mrad, the pt0 w_η^{CoG} still improves (1.6×), but at 1.5 mrad it degrades (4.4×), indicating that the crossover pileup fraction for w_η^{CoG} lies between 7.9% and 22.4% at low E_T.

6.2 Crossing-angle comparison

The improvement factors scale with the pileup fraction: the 22.4% admixture at 0 mrad yields larger improvements (up to 20.8× for w_η^{CoG} at pt3) than the 7.9% admixture at 1.5 mrad (up to 16.1×). Conversely, the 1.5 mrad nominals are already much better than their 0 mrad counterparts (e.g., BDT pt1: 1.16 vs. 12.20), confirming that lower pileup naturally reduces the data–MC discrepancy.

6.3 Low-E_T degradation mechanism

The low-E_T degradation affects the BDT score most severely because the BDT amplifies subtle shower shape distortions through its nonlinear decision function. Contributing factors include:

1. **Shared MC samples**: the same `photon10_double` and `jet12_double` samples are used for both crossing angles; beam conditions in the MC may better match one period than the other.

2. **Low-E_T sensitivity**: clusters near the 8–14 GeV threshold are most sensitive to vertex shifts and energy redistribution.

3. **Small-fraction noise**: at 7.9% mixing, any mismatch between the double MC and real pileup gets amplified relative to the small physical correction.

7 Conclusions and Recommendations

1. **0 mrad period**: the 22.4% double-interaction blending substantially improves data–MC agreement for pileup-sensitive variables (w_η^{CoG}, BDT, f_1) at $E_T \geq 14\,\mathrm{GeV}$, with improvement factors up to 20.8×. At pt0 (10–14 GeV), the BDT score degrades by 3.0× while w_η^{CoG} still improves modestly.

2. **1.5 mrad period**: the 7.9% blending helps at high E_T ($\geq 18\,\mathrm{GeV}$) but degrades both w_η^{CoG} and BDT at low E_T. The nominal agreement is already acceptable ($\chi^2/\mathrm{ndf} \lesssim 10$ for most variables).

3. **Recommended baseline for cross-section measurement**:

   - **0 mrad,** $E_T \geq 14\,\mathbf{GeV}$: use mixed MC (22.4% DI)
   - **0 mrad,** $E_T < 14\,\mathbf{GeV}$: use mixed MC for w_η^{CoG}; use nominal MC for BDT-sensitive quantities; assign nominal–mixed difference as systematic
   - **1.5 mrad,** $E_T \geq 18\,\mathbf{GeV}$: use mixed MC (7.9% DI)
   - **1.5 mrad,** $E_T < 18\,\mathbf{GeV}$: use nominal MC; assign nominal–mixed difference as systematic

4. **Alternative**: use nominal MC for the full 1.5 mrad period and apply the double-interaction correction to 0 mrad data only.